BCSC0805: DATA STRUCTURES & ALGORITHMS LAB

Objective: *The objective of this course is that students will understand and implement simple data structures, able demonstrate different sorting and searching techniques. and will be familiar with graphs and their applications.*

CREDITS:01 **L-T-P-J:0-0-2-0**

Module	Content	Lab Hours
I	• Program to implement various operations in a singly linked list. • Program to implement insertion, deletion and traversal in a doubly linked List. • Program to implement polynomial addition using linked list. • Program to demonstrate the various operations on stack. • Program to convert an infix expression into postfix expression. • Program to evaluate a given postfix expression. • Program to implement Tower of Hanoi problem using Recursion.	12
II	• Program to demonstrate the implementation of various operations on linear and circular queue. • Program to demonstrate the implementation of insertion and traversals on a binary search tree. • Program to implement Dijkstra's Algorithm to find the shortest path between source and destination. • Program to search a given element as entered by the user using sequential and binary search to search a given element as entered by the user. • Implementation of various sorting algorithms like Selection Sort, Bubble Sort, Insertion Sort, Merge Sort, Quick Sort and Heap Sort.	12

Note: All Code must be done in Java as well as Python

Text Book:

- Aaron M. Tanenbaum, YedidyahLangsam and Moshe J. Augenstein, "Data Structures Using C and C++", 2nd Edition, PHI, 2009.

Reference Books:

- Horowitz and Sahani, "Fundamentals of Data Structures", 3rd Edition, W H Freeman & Co, 2004- 05.
- Jean Paul Trembley and Paul G. Sorenson, "An Introduction to Data Structures with Applications", 2nd Edition, TMH, 2007.
- R. Kruse, "Data Structures and Program Design in C", 2nd Edition, Pearson Education, 2004.
- Lipschutz Schaum's Outline Series, "Data Structures", 12th Reprint, TMH, 2010.
- G A V Pai, "Data Structures and Algorithms", TMH, 2009.

Outcome: After completion of course, student will be able to:

- CO1: Demonstrate the associated operations in linear data structure like stack, Queue and link list.
- CO2: Demonstrate the associated operations in Binary Search Tree and Dijkstra's Algorithm.
- CO3: Implementation the sorting algorithms like Selection Sort, Bubble Sort, Insertion Sort, Merge Sort, Quick Sort and Heap Sort.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1/PSO1
CO2	PO4/PSO1,PSO3
CO3	PO2/PSO3,PSO4